

CALARASI, Romania

WMO: **154600**Lat: **44.2058N**Lon: **27.3383E**Elev: **20**StdP: **101.09**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.3	-9.4	-15.0	1.0	-11.8	-12.0	1.3	-8.4	11.4	-3.3	9.7	-3.4	2.5	20	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	34.7	21.3	33.1	20.8	31.5	20.4	23.3	30.3	22.4	29.4	21.6	28.5	2.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.3	16.0	26.3	20.3	15.0	25.1	19.4	14.2	24.2	69.5	30.2	65.9	29.4	62.9	28.4	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 8.0	6.7	5.7		-16.5	37.2	3.2	2.2	-18.8	38.8	-20.7	40.1	-22.5	41.3	-24.9	42.9	
(5)			DB	-16.8	24.7	3.1	1.5	-19.0	25.8	-20.8	26.7	-22.6	27.6	-24.8	28.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	-0.3	2.4	7.2	12.5	17.8	22.0	24.1	23.7	18.9	12.7	7.3	1.5
	DBStd	9.41	5.23	5.55	4.63	3.78	3.39	3.02	2.60	2.78	3.50	4.23	4.62	4.77
	HDD10.0	1058	319	216	112	18	1	0	0	0	22	107	264	264
	HDD18.3	2676	577	446	346	178	52	6	1	36	181	332	521	521
	CDD10.0	1982	1	3	24	92	241	359	437	426	267	104	25	2
	CDD18.3	558	0	0	0	3	34	115	179	168	53	6	1	0
	CDH23.3	5900	0	0	3	56	388	1162	1886	1824	519	59	3	0
(13)	CDH26.7	2423	0	0	0	10	91	453	854	842	164	9	0	0
(14) Wind	WSAvg	2.4	2.7	2.8	2.9	2.7	2.3	2.1	2.0	2.0	2.2	2.1	2.3	2.5
(15) Precipitation	PrecAvg	492	31	30	31	36	58	64	49	47	38	33	39	37
	PrecMax	809	147	86	96	64	172	132	114	144	119	140	118	149
	PrecMin	318	5	2	0	0	8	15	8	2	0	0	1	5
	PrecStd	112	28	19	23	16	38	32	27	35	34	27	27	31
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	18.8	23.0	28.1	31.2	35.0	37.5	37.1	32.8	27.7	23.0	16.4
		MCWB	9.9	11.9	12.7	16.9	19.2	21.9	21.1	21.3	19.3	18.2	15.6	11.9
	2%	DB	11.0	15.5	20.2	24.7	29.0	32.7	34.8	34.6	30.5	24.8	19.6	13.1
		MCWB	7.7	10.0	12.1	15.4	18.4	20.9	21.1	21.4	18.7	16.9	14.4	9.6
	5%	DB	8.6	12.5	17.8	22.1	27.3	31.1	33.0	33.0	28.5	22.3	16.6	10.2
		MCWB	6.4	8.5	11.0	14.2	17.9	20.6	21.0	20.9	18.3	15.7	13.0	8.0
	10%	DB	6.4	9.9	15.2	19.9	25.5	29.5	31.2	31.3	26.4	19.8	14.2	7.9
		MCWB	5.0	7.1	9.6	13.0	17.0	20.1	20.6	20.2	17.7	14.6	11.6	6.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	12.5	13.9	17.7	21.1	23.9	24.2	25.4	21.5	19.3	16.2	12.1
		MCDB	13.3	17.7	20.8	26.1	28.4	31.0	31.4	32.7	28.1	25.2	21.8	15.6
	2%	WB	8.2	10.5	12.5	16.2	19.8	22.7	23.1	23.5	20.2	17.9	14.7	10.0
		MCDB	10.5	14.6	19.0	23.6	26.7	29.9	30.4	30.0	26.3	23.0	18.7	12.6
	5%	WB	6.6	8.9	11.3	14.8	18.8	21.7	22.3	22.2	19.3	16.5	13.4	8.2
		MCDB	8.4	12.1	17.3	20.5	25.3	28.8	29.4	28.8	25.8	21.1	16.3	10.1
	10%	WB	5.0	7.2	9.9	13.7	17.8	20.8	21.6	21.2	18.5	15.2	11.9	6.4
		MCDB	6.6	9.8	14.5	18.6	23.7	27.5	28.4	28.3	24.8	19.1	14.1	7.9
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	8.1	10.0	11.6	12.3	12.1	12.5	13.2	12.2	10.5	8.1	6.7
		MCDBR	9.9	13.0	14.8	15.3	15.3	14.8	15.5	15.9	15.6	14.6	12.8	10.9
	5% WB	MCWBR	7.0	8.2	7.9	7.6	6.7	5.4	5.0	5.2	5.9	7.3	7.7	7.8
		MCDBR	9.5	12.5	13.9	13.4	13.6	13.3	13.0	13.3	13.7	13.2	11.9	10.4
		MCWBR	7.1	8.2	7.7	7.0	6.4	5.4	5.0	5.1	5.7	6.9	7.3	7.6
(40) Clear-Sky Solar Irradiance	taub	0.347	0.377	0.411	0.461	0.436	0.441	0.453	0.466	0.435	0.395	0.369	0.349	0.349
	taud	2.290	2.218	2.174	2.044	2.199	2.218	2.195	2.151	2.205	2.299	2.320	2.309	2.309
	Ebn at Noon	751	792	815	802	839	834	817	790	782	764	719	707	707
	Edh at Noon	85	109	130	160	141	139	141	142	125	100	83	77	77
(44) All-Sky Solar Radiation	RadAvg	1.38	2.18	3.35	4.72	5.95	6.46	6.53	5.84	4.25	2.70	1.52	1.17	1.17
	RadStd	0.20	0.32	0.34	0.49	0.41	0.44	0.40	0.32	0.32	0.29	0.19	0.15	0.15

Historical Trends

Station Only	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	-119	+87	+55